

Variational Inequality

Find $\bar{x} \in K$ such that

$$(x - \bar{x})^T F(\bar{x}) \geq 0$$

$$\forall x \in K$$

Equivalent

Fixed Point Problem

Find $\bar{x} \in K$ such that

$$\bar{x} = P_K(\bar{x} - \alpha F(\bar{x}))$$

$$\alpha > 0$$

Normal Equation Form

$$0 \in F(\bar{x}) + N_K(\bar{x})$$